Ordering of Unit Roots

1.Convention $\omega_n^0 = \omega_n^n = 1$

1.

$$\omega_2^1 = -1$$

2.

$$\omega_3^1 = \frac{-1+\sqrt{3}i}{2}$$

$$\omega_3^2 = \frac{-1-\sqrt{3}i}{2}$$

3.

$$\omega_4^1 = i$$

$$\omega_4^2 = \omega_2^1$$

$$\omega_4^3 = -i$$

4.

$$\omega_5^1 = \frac{-1+\sqrt{5}+\sqrt{10+\sqrt{5}}i}{4}$$

$$\omega_5^2 = \frac{-1+\sqrt{5}+\sqrt{10-\sqrt{5}}i}{4}$$

$$\omega_5^3 = \frac{-1-\sqrt{5}-\sqrt{10-\sqrt{5}}i}{4}$$

$$\omega_5^4 = \frac{-1+\sqrt{5}-\sqrt{10+\sqrt{5}}i}{4}$$

5.

$$\omega_6^1 = \frac{1+\sqrt{3}i}{2}$$

$$\omega_6^2 = \omega_3^1$$

$$\omega_6^3 = \omega_2^1$$

$$\omega_6^4 = \omega_3^2$$

$$\omega_6^5 = \frac{1-\sqrt{3}i}{2}$$

6.

$$\vec{\mathrm{A}}_7 = \sqrt[3]{7}\begin{bmatrix} \sqrt[3]{\frac{1+3\sqrt{3}i}{2}} \\ \sqrt[3]{\frac{1-3\sqrt{3}i}{2}} \end{bmatrix}$$

$$\vec{\Omega}_3^0 = \begin{bmatrix} 1 & 1 \end{bmatrix}$$

$$\vec{\Omega}_3^1 = \begin{bmatrix} \omega_3^1 & \omega_3^2 \end{bmatrix}$$

$$\vec{\Omega}_3^2 = \begin{bmatrix} \omega_3^2 & \omega_3^1 \end{bmatrix}$$

$$\omega_7^1 = \frac{-1 + \vec{\Omega}_3^0 \cdot \vec{\mathrm{A}}_7 + \sqrt{21 - 3\vec{\Omega}_3^2 \cdot \vec{\mathrm{A}}_7}i}{6}$$

$$\omega_7^2 = \frac{-1 + \vec{\Omega}_3^2 \cdot \vec{\mathrm{A}}_7 + \sqrt{21 - 3\vec{\Omega}_3^1 \cdot \vec{\mathrm{A}}_7}i}{6}$$

$$\omega_7^3 = \frac{-1 + \vec{\Omega}_3^1 \cdot \vec{\mathrm{A}}_7 + \sqrt{21 - 3\vec{\Omega}_3^0 \cdot \vec{\mathrm{A}}_7}i}{6}$$

$$\omega_7^4 = \frac{-1 + \vec{\Omega}_3^1 \cdot \vec{\mathrm{A}}_7 - \sqrt{21 - 3\vec{\Omega}_3^0 \cdot \vec{\mathrm{A}}_7}i}{6}$$

$$\omega_7^5 = \frac{-1 + \vec{\Omega}_3^2 \cdot \vec{\mathrm{A}}_7 - \sqrt{21 - 3\vec{\Omega}_3^1 \cdot \vec{\mathrm{A}}_7}i}{6}$$

$$\omega_7^6 = \frac{-1 + \vec{\Omega}_3^0 \cdot \vec{\mathrm{A}}_7 - \sqrt{21 - 3\vec{\Omega}_3^2 \cdot \vec{\mathrm{A}}_7}i}{6}$$

7.

$$\omega_8^1 = \frac{\sqrt{2} + \sqrt{2}i}{2}$$

$$\omega_8^2 = \omega_4^1$$

$$\omega_8^3 = \frac{-\sqrt{2} + \sqrt{2}i}{2}$$

$$\omega_8^4 = \omega_2^1$$

$$\omega_8^5 = \frac{-\sqrt{2} - \sqrt{2}i}{2}$$

$$\omega_8^6 = \omega_4^3$$

$$\omega_8^7 = \frac{\sqrt{2} - \sqrt{2}i}{2}$$

8.

$$\omega_9^1 = \sqrt[3]{\omega_3^1}$$

$$\omega_9^2 = \omega_3^1 \sqrt[3]{\omega_3^2}$$

$$\omega_9^3 = \omega_3^1$$

$$\omega_9^4 = \omega_3^1 \sqrt[3]{\omega_3^1}$$

$$\omega_9^5 = \omega_3^2 \sqrt[3]{\omega_3^2}$$

$$\omega_9^6 = \omega_3^2$$

$$\omega_9^7 = \omega_3^2 \sqrt[3]{\omega_3^1}$$

$$\omega_9^8 = \sqrt[3]{\omega_3^2}$$

10.

$$\omega_{10}^1 = \frac{1+\sqrt{5}+\sqrt{10-2\sqrt{5}}i}{4}$$

$$\omega_{10}^2 = \omega_5^1$$

$$\omega_{10}^3 = \frac{1-\sqrt{5}+\sqrt{10+2\sqrt{5}}i}{4}$$

$$\omega_{10}^4 = \omega_5^2$$

$$\omega_{10}^5 = \omega_2^1$$

$$\omega_{10}^6 = \omega_5^3$$

$$\omega_{10}^7 = \frac{1-\sqrt{5}-\sqrt{10+2\sqrt{5}}i}{4}$$

$$\omega_{10}^8 = \omega_5^4$$

$$\omega_{10}^9 = \frac{1+\sqrt{5}-\sqrt{10-2\sqrt{5}}i}{4}$$

11. $X_{11}^1 = \frac{-178+50\sqrt{5}+(65+25\sqrt{5})\sqrt{10-2\sqrt{5}}i}{8}$

$$X_{11}^2 = \frac{-178-50\sqrt{5}+(65-25\sqrt{5})\sqrt{10+2\sqrt{5}}i}{8}$$

$$\vec{\mathrm{A}}_{11} = \sqrt[5]{11}\begin{bmatrix} \sqrt[5]{X_{11}^1} \\ \sqrt[5]{\bar{X}_{11}^1} \\ \sqrt[5]{X_{11}^2} \\ \sqrt[5]{\bar{X}_{11}^2} \end{bmatrix}$$

$$\vec{\Omega}_5^0 = \begin{bmatrix} 1 & 1 & \omega_5^2 & \omega_5^3 \end{bmatrix}$$

$$\vec{\Omega}_5^1 = \begin{bmatrix} \omega_5^1 & \omega_5^4 & 1 & 1 \end{bmatrix}$$

$$\vec{\Omega}_5^2 = \begin{bmatrix} \omega_5^2 & \omega_5^3 & \omega_5^3 & \omega_5^2 \end{bmatrix}$$

$$\vec{\Omega}_5^3 = \begin{bmatrix} \omega_5^3 & \omega_5^2 & \omega_5^1 & \omega_5^4 \end{bmatrix}$$

$$\vec{\Omega}_5^4 = \begin{bmatrix} \omega_5^4 & \omega_5^1 & \omega_5^4 & \omega_5^1 \end{bmatrix}$$

$$\omega_{11}^1 = \frac{-1+\vec{\Omega}_5^4 \cdot \vec{\mathrm{A}}_{11} + \sqrt{55-5\vec{\Omega}_5^1 \cdot \vec{\mathrm{A}}_{11}}i}{10}$$

$$\omega_{11}^{2} = \frac{-1 + \vec{\Omega}_{5}^{1} \cdot \vec{\mathrm{A}}_{11} + \sqrt{55 - 5\vec{\Omega}_{5}^{3} \cdot \vec{\mathrm{A}}_{11}}i}{10}$$

$$\omega_{11}^{3} = \frac{-1 + \vec{\Omega}_{5}^{0} \cdot \vec{\mathrm{A}}_{11} + \sqrt{55 - 5\vec{\Omega}_{5}^{2} \cdot \vec{\mathrm{A}}_{11}}i}{10}$$

$$\omega_{11}^{4} = \frac{-1 + \vec{\Omega}_{5}^{3} \cdot \vec{\mathrm{A}}_{11} + \sqrt{55 - 5\vec{\Omega}_{5}^{0} \cdot \vec{\mathrm{A}}_{11}}i}{10}$$

$$\omega_{11}^{5} = \frac{-1 + \vec{\Omega}_{5}^{2} \cdot \vec{\mathrm{A}}_{11} + \sqrt{55 - 5\vec{\Omega}_{5}^{4} \cdot \vec{\mathrm{A}}_{11}}i}{10}$$

$$\omega_{11}^{6} = \frac{-1 + \vec{\Omega}_{5}^{2} \cdot \vec{\mathrm{A}}_{11} - \sqrt{55 - 5\vec{\Omega}_{5}^{4} \cdot \vec{\mathrm{A}}_{11}}i}{10}$$

$$\omega_{11}^{7} = \frac{-1 + \vec{\Omega}_{5}^{3} \cdot \vec{\mathrm{A}}_{11} - \sqrt{55 - 5\vec{\Omega}_{5}^{0} \cdot \vec{\mathrm{A}}_{11}}i}{10}$$

$$\omega_{11}^{8} = \frac{-1 + \vec{\Omega}_{5}^{0} \cdot \vec{\mathrm{A}}_{11} - \sqrt{55 - 5\vec{\Omega}_{5}^{2} \cdot \vec{\mathrm{A}}_{11}}i}{10}$$

$$\omega_{11}^{9} = \frac{-1 + \vec{\Omega}_{5}^{1} \cdot \vec{\mathrm{A}}_{11} - \sqrt{55 - 5\vec{\Omega}_{5}^{3} \cdot \vec{\mathrm{A}}_{11}}i}{10}$$

$$\omega_{11}^{10} = \frac{-1 + \vec{\Omega}_{5}^{4} \cdot \vec{\mathrm{A}}_{11} - \sqrt{55 - 5\vec{\Omega}_{5}^{1} \cdot \vec{\mathrm{A}}_{11}}i}{10}$$

12.

$$\omega_{12}^{1} = \frac{\sqrt{3} + i}{2}$$

$$\omega_{12}^{2} = \omega_{6}^{1}$$

$$\omega_{12}^{3} = \omega_{4}^{1}$$

$$\omega_{12}^{4} = \omega_{3}^{1}$$

$$\omega_{12}^{5} = \frac{-\sqrt{3} + i}{2}$$

$$\omega_{12}^{6} = \omega_{2}^{1}$$

$$\omega_{12}^{7} = \frac{-\sqrt{3} - i}{2}$$

$$\omega_{12}^{8} = \omega_{3}^{2}$$

$$\omega_{12}^{9} = \omega_{4}^{3}$$

$$\omega_{12}^{10} = \omega_{6}^{5}$$

$$\omega_{12}^{11} = \frac{\sqrt{3} - i}{2}$$

13.

$$\vec{\mathrm{A}}_{13} = \sqrt[3]{13} \begin{bmatrix} \sqrt[3]{\frac{-5+3\sqrt{3}i}{2}} \\ \sqrt[3]{\frac{-5-3\sqrt{3}i}{2}} \end{bmatrix}$$

$$\vec{\Lambda}_{13}^{1} = \begin{bmatrix} \frac{1+3\sqrt{3}i}{2} & \frac{1-3\sqrt{3}i}{2} \end{bmatrix}$$

$$\vec{\Lambda}_{13}^{2} = \begin{bmatrix} \frac{-5-\sqrt{3}i}{2} & \frac{-5+\sqrt{3}i}{2} \end{bmatrix}$$

$$\vec{\Lambda}_{13}^{3} = \begin{bmatrix} 2-\sqrt{3}i & 2+\sqrt{3}i \end{bmatrix}$$

$$\omega_{13}^{1} = \frac{-1+\vec{\Omega}_{3}^{2}\cdot\vec{\mathrm{A}}_{13}+\sqrt{39+3\vec{\Lambda}_{13}^{3}\cdot\vec{\mathrm{A}}_{13}}+\sqrt{78-6\vec{\Omega}_{3}^{0}\cdot\vec{\mathrm{A}}_{13}-6\sqrt{39+3\vec{\Lambda}_{13}^{1}\cdot\vec{\mathrm{A}}_{13}}i}}{12}$$

$$\omega_{13}^{2} = \frac{-1+\vec{\Omega}_{3}^{0}\cdot\vec{\mathrm{A}}_{13}+\sqrt{39+3\vec{\Lambda}_{13}^{1}\cdot\vec{\mathrm{A}}_{13}}+\sqrt{78-6\vec{\Omega}_{3}^{1}\cdot\vec{\mathrm{A}}_{13}-6\sqrt{39+3\vec{\Lambda}_{13}^{2}\cdot\vec{\mathrm{A}}_{13}}i}}{12}$$

$$\omega_{13}^{3} = \frac{-1+\vec{\Omega}_{3}^{0}\cdot\vec{\mathrm{A}}_{13}-\sqrt{39+3\vec{\Lambda}_{13}^{3}\cdot\vec{\mathrm{A}}_{13}}+\sqrt{78-6\vec{\Omega}_{3}^{1}\cdot\vec{\mathrm{A}}_{13}+6\sqrt{39+3\vec{\Lambda}_{13}^{2}\cdot\vec{\mathrm{A}}_{13}}i}}{12}$$

$$\omega_{13}^{4} = \frac{-1+\vec{\Omega}_{3}^{1}\cdot\vec{\mathrm{A}}_{13}+\sqrt{39+3\vec{\Lambda}_{13}^{2}\cdot\vec{\mathrm{A}}_{13}}+\sqrt{78-6\vec{\Omega}_{3}^{2}\cdot\vec{\mathrm{A}}_{13}+6\sqrt{39+3\vec{\Lambda}_{13}^{3}\cdot\vec{\mathrm{A}}_{13}}i}}{12}$$

$$\omega_{13}^{5} = \frac{-1+\vec{\Omega}_{3}^{2}\cdot\vec{\mathrm{A}}_{13}-\sqrt{39+3\vec{\Lambda}_{13}^{3}\cdot\vec{\mathrm{A}}_{13}}+\sqrt{78-6\vec{\Omega}_{3}^{0}\cdot\vec{\mathrm{A}}_{13}+6\sqrt{39+3\vec{\Lambda}_{13}^{1}\cdot\vec{\mathrm{A}}_{13}}i}}{12}$$

$$\omega_{13}^{6} = \frac{-1+\vec{\Omega}_{3}^{1}\cdot\vec{\mathrm{A}}_{13}-\sqrt{39+3\vec{\Lambda}_{13}^{2}\cdot\vec{\mathrm{A}}_{13}}+\sqrt{78-6\vec{\Omega}_{3}^{2}\cdot\vec{\mathrm{A}}_{13}-6\sqrt{39+3\vec{\Lambda}_{13}^{3}\cdot\vec{\mathrm{A}}_{13}}i}}{12}$$

$$\omega_{13}^{7} = \frac{-1+\vec{\Omega}_{3}^{1}\cdot\vec{\mathrm{A}}_{13}-\sqrt{39+3\vec{\Lambda}_{13}^{2}\cdot\vec{\mathrm{A}}_{13}}-\sqrt{78-6\vec{\Omega}_{3}^{2}\cdot\vec{\mathrm{A}}_{13}-6\sqrt{39+3\vec{\Lambda}_{13}^{3}\cdot\vec{\mathrm{A}}_{13}}i}}{12}$$

$$\omega_{13}^{8} = \frac{-1+\vec{\Omega}_{3}^{2}\cdot\vec{\mathrm{A}}_{13}-\sqrt{39+3\vec{\Lambda}_{13}^{3}\cdot\vec{\mathrm{A}}_{13}}-\sqrt{78-6\vec{\Omega}_{3}^{0}\cdot\vec{\mathrm{A}}_{13}-6\sqrt{39+3\vec{\Lambda}_{13}^{1}\cdot\vec{\mathrm{A}}_{13}}i}}{12}$$

$$\omega_{13}^{9} = \frac{-1+\vec{\Omega}_{3}^{1}\cdot\vec{\mathrm{A}}_{13}+\sqrt{39+3\vec{\Lambda}_{13}^{2}\cdot\vec{\mathrm{A}}_{13}}-\sqrt{78-6\vec{\Omega}_{3}^{2}\cdot\vec{\mathrm{A}}_{13}+6\sqrt{39+3\vec{\Lambda}_{13}^{3}\cdot\vec{\mathrm{A}}_{13}}i}}{12}$$

$$\omega_{13}^{10} = \frac{-1+\vec{\Omega}_{3}^{0}\cdot\vec{\mathrm{A}}_{13}-\sqrt{39+3\vec{\Lambda}_{13}^{3}\cdot\vec{\mathrm{A}}_{13}}-\sqrt{78-6\vec{\Omega}_{3}^{1}\cdot\vec{\mathrm{A}}_{13}+6\sqrt{39+3\vec{\Lambda}_{13}^{2}\cdot\vec{\mathrm{A}}_{13}}i}}{12}$$

$$\omega_{13}^{11} = \frac{-1+\vec{\Omega}_{3}^{0}\cdot\vec{\mathrm{A}}_{13}+\sqrt{39+3\vec{\Lambda}_{13}^{1}\cdot\vec{\mathrm{A}}_{13}}-\sqrt{78-6\vec{\Omega}_{3}^{1}\cdot\vec{\mathrm{A}}_{13}-6\sqrt{39+3\vec{\Lambda}_{13}^{2}\cdot\vec{\mathrm{A}}_{13}}i}}{12}$$

$$\omega_{13}^{12} = \frac{-1+\vec{\Omega}_{3}^{2}\cdot\vec{\mathrm{A}}_{13}+\sqrt{39+3\vec{\Lambda}_{13}^{3}\cdot\vec{\mathrm{A}}_{13}}-\sqrt{78-6\vec{\Omega}_{3}^{0}\cdot\vec{\mathrm{A}}_{13}-6\sqrt{39+3\vec{\Lambda}_{13}^{1}\cdot\vec{\mathrm{A}}_{13}}i}}{12}$$

1.

$$\omega_{14}^{1} = -\omega_{7}^{4}$$
$$\omega_{14}^{2} = \omega_{7}^{1}$$
$$\omega_{14}^{3} = -\omega_{7}^{5}$$
$$\omega_{14}^{4} = \omega_{7}^{2}$$

$$\omega_{14}^{5} = -\omega_{7}^{6}$$
$$\omega_{14}^{6} = \omega_{7}^{3}$$
$$\omega_{14}^{7} = \omega_{2}^{1}$$
$$\omega_{14}^{8} = \omega_{7}^{4}$$

$$\omega_{14}^{9} = -\omega_{7}^{1}$$
$$\omega_{14}^{10} = \omega_{7}^{5}$$
$$\omega_{14}^{11} = -\omega_{7}^{2}$$
$$\omega_{14}^{12} = \omega_{7}^{6}$$

$$\omega_{14}^{13} = -\omega_{7}^{3}$$

2.

$$\omega_{15}^{1} = \frac{1+\sqrt{5}+\sqrt{30-6\sqrt{5}}+(\sqrt{3}+\sqrt{15}-\sqrt{10-2\sqrt{5}})i}{8}$$

$$\omega_{15}^{2} = \frac{1-\sqrt{5}+\sqrt{30+6\sqrt{5}}+(-\sqrt{3}+\sqrt{15}+\sqrt{10+2\sqrt{5}})i}{8}$$

$$\omega_{15}^{3} = \omega_{5}^{1}$$

$$\omega_{15}^{4} = \frac{1+\sqrt{5}-\sqrt{30-6\sqrt{5}}+(\sqrt{3}+\sqrt{15}+\sqrt{10-2\sqrt{5}})i}{8}$$

$$\omega_{15}^{5} = \omega_{3}^{1}$$

$$\omega_{15}^{6} = \omega_{5}^{2}$$

$$\omega_{15}^{7} = \frac{1-\sqrt{5}-\sqrt{30+6\sqrt{5}}+(\sqrt{3}-\sqrt{15}+\sqrt{10+2\sqrt{5}})i}{8}$$

$$\omega_{15}^{8} = \frac{1-\sqrt{5}-\sqrt{30+6\sqrt{5}}-(\sqrt{3}-\sqrt{15}+\sqrt{10+2\sqrt{5}})i}{8}$$

$$\omega_{15}^{9} = \omega_{5}^{3}$$

$$\omega_{15}^{10} = \omega_{3}^{2}$$

$$\omega_{15}^{11} = \frac{1+\sqrt{5}-\sqrt{30-6\sqrt{5}}-(\sqrt{3}+\sqrt{15}+\sqrt{10-2\sqrt{5}})i}{8}$$

$$\omega_{15}^{12} = \omega_5^4$$

$$\omega_{15}^{13} = \frac{1-\sqrt{5}+\sqrt{30+6\sqrt{5}}-(-\sqrt{3}+\sqrt{15}+\sqrt{10+2\sqrt{5}})i}{8}$$

$$\omega_{15}^{14} = \frac{1+\sqrt{5}+\sqrt{30-6\sqrt{5}}-(\sqrt{3}+\sqrt{15}-\sqrt{10-2\sqrt{5}})i}{8}$$

3.

$$\omega_{16}^{1} = \frac{\sqrt{2+\sqrt{2}}+\sqrt{2-\sqrt{2}}i}{2}$$

$$\omega_{16}^{2} = \omega_8^1$$

$$\omega_{16}^{3} = \frac{\sqrt{2-\sqrt{2}}+\sqrt{2+\sqrt{2}}i}{2}$$

$$\omega_{16}^{4} = \omega_4^1$$

$$\omega_{16}^{5} = \frac{-\sqrt{2-\sqrt{2}}+\sqrt{2+\sqrt{2}}i}{2}$$

$$\omega_{16}^{6} = \omega_8^3$$

$$\omega_{16}^{7} = \frac{-\sqrt{2+\sqrt{2}}+\sqrt{2-\sqrt{2}}i}{2}$$

$$\omega_{16}^{8} = \omega_2^1$$

$$\omega_{16}^{9} = \frac{-\sqrt{2+\sqrt{2}}-\sqrt{2-\sqrt{2}}i}{2}$$

$$\omega_{16}^{10} = \omega_8^5$$

$$\omega_{16}^{11} = \frac{-\sqrt{2-\sqrt{2}}-\sqrt{2+\sqrt{2}}i}{2}$$

$$\omega_{16}^{12} = \omega_4^3$$

$$\omega_{16}^{13} = \frac{\sqrt{2-\sqrt{2}}-\sqrt{2+\sqrt{2}}i}{2}$$

$$\omega_{16}^{14} = \omega_8^7$$

$$\omega_{16}^{15} = \frac{\sqrt{2+\sqrt{2}}-\sqrt{2-\sqrt{2}}i}{2}$$

4.

$$X_{17}^1 = 17 + 4\sqrt{17}$$
$$X_{17}^2 = 17 - 4\sqrt{17}$$

$$Y_{17}^1 = 13 + 3\sqrt{17}$$
$$Y_{17}^2 = 13 - 3\sqrt{17}$$

$$\Lambda_{17}^1 = 5 - \sqrt{17}$$
$$\Lambda_{17}^2 = 5 + \sqrt{17}$$

18.

$$\omega_{17}^1 = \frac{-1 + \sqrt{17} + \Lambda_{17}^1\sqrt{X_{17}^1} + 2\sqrt{4 + Y_{17}^1(1 - \sqrt{X_{17}^2})} + 2\sqrt{34 - 2\sqrt{17} + 2\Lambda_{17}^1\sqrt{X_{17}^1} + 4\sqrt{4 + Y_{17}^1(1 + \sqrt{X_{17}^2})}}i}{16}$$

$$\omega_{17}^2 = \frac{-1 + \sqrt{17} - \Lambda_{17}^1\sqrt{X_{17}^1} + 2\sqrt{4 + Y_{17}^1(1 + \sqrt{X_{17}^2})} + 2\sqrt{34 - 2\sqrt{17} - 2\Lambda_{17}^1\sqrt{X_{17}^1} + 4\sqrt{4 + Y_{17}^1(1 - \sqrt{X_{17}^2})}}i}{16}$$

$$\omega_{17}^3 = \frac{-1 + \sqrt{17} + \Lambda_{17}^2\sqrt{X_{17}^2} + 2\sqrt{4 + Y_{17}^2(1 + \sqrt{X_{17}^1})} + 2\sqrt{34 + 2\sqrt{17} + 2\Lambda_{17}^2\sqrt{X_{17}^2} - 4\sqrt{4 + Y_{17}^2(1 - \sqrt{X_{17}^1})}}i}{16}$$

$$\omega_{17}^4 = \frac{-1 + \sqrt{17} + \Lambda_{17}^1\sqrt{X_{17}^1} - 2\sqrt{4 + Y_{17}^1(1 - \sqrt{X_{17}^2})} - 2\sqrt{34 - 2\sqrt{17} + 2\Lambda_{17}^1\sqrt{X_{17}^1} - 4\sqrt{4 + Y_{17}^1(1 + \sqrt{X_{17}^2})}}i}{16}$$

$$\omega_{17}^5 = \frac{-1 + \sqrt{17} + \Lambda_{17}^2\sqrt{X_{17}^2} - 2\sqrt{4 + Y_{17}^2(1 + \sqrt{X_{17}^1})} + 2\sqrt{34 + 2\sqrt{17} + 2\Lambda_{17}^2\sqrt{X_{17}^2} + 4\sqrt{4 + Y_{17}^2(1 - \sqrt{X_{17}^1})}}i}{16}$$

$$\omega_{17}^6 = \frac{-1 + \sqrt{17} - \Lambda_{17}^2\sqrt{X_{17}^2} + 2\sqrt{4 + Y_{17}^2(1 - \sqrt{X_{17}^1})} + 2\sqrt{34 + 2\sqrt{17} - 2\Lambda_{17}^2\sqrt{X_{17}^2} + 4\sqrt{4 + Y_{17}^2(1 + \sqrt{X_{17}^1})}}i}{16}$$

$$\omega_{17}^7 = \frac{-1 + \sqrt{17} - \Lambda_{17}^2\sqrt{X_{17}^2} - 2\sqrt{4 + Y_{17}^2(1 - \sqrt{X_{17}^1})} + 2\sqrt{34 + 2\sqrt{17} - 2\Lambda_{17}^2\sqrt{X_{17}^2} - 4\sqrt{4 + Y_{17}^2(1 + \sqrt{X_{17}^1})}}i}{16}$$

$$\omega_{17}^8 = \frac{-1 + \sqrt{17} - \Lambda_{17}^1\sqrt{X_{17}^1} - 2\sqrt{4 + Y_{17}^1(1 + \sqrt{X_{17}^2})} + 2\sqrt{34 - 2\sqrt{17} - 2\Lambda_{17}^1\sqrt{X_{17}^1} - 4\sqrt{4 + Y_{17}^1(1 - \sqrt{X_{17}^2})}}i}{16}$$

$$\omega_{17}^9 = \frac{-1 + \sqrt{17} - \Lambda_{17}^1\sqrt{X_{17}^1} - 2\sqrt{4 + Y_{17}^1(1 + \sqrt{X_{17}^2})} - 2\sqrt{34 - 2\sqrt{17} - 2\Lambda_{17}^1\sqrt{X_{17}^1} - 4\sqrt{4 + Y_{17}^1(1 - \sqrt{X_{17}^2})}}i}{16}$$

$$\omega_{17}^{10} = \frac{-1 + \sqrt{17} - \Lambda_{17}^2\sqrt{X_{17}^2} - 2\sqrt{4 + Y_{17}^2(1 - \sqrt{X_{17}^1})} - 2\sqrt{34 + 2\sqrt{17} - 2\Lambda_{17}^2\sqrt{X_{17}^2} - 4\sqrt{4 + Y_{17}^2(1 + \sqrt{X_{17}^1})}}i}{16}$$

$$\omega_{17}^{11} = \frac{-1 + \sqrt{17} - \Lambda_{17}^2\sqrt{X_{17}^2} + 2\sqrt{4 + Y_{17}^2(1 - \sqrt{X_{17}^1})} - 2\sqrt{34 + 2\sqrt{17} - 2\Lambda_{17}^2\sqrt{X_{17}^2} + 4\sqrt{4 + Y_{17}^2(1 + \sqrt{X_{17}^1})}}i}{16}$$

$$\omega_{17}^{12} = \frac{-1+\sqrt{17}+\Lambda_{17}^2\sqrt{X_{17}^2}-2\sqrt{4+Y_{17}^2(1+\sqrt{X_{17}^1})}-2\sqrt{34+2\sqrt{17}+2\Lambda_{17}^2\sqrt{X_{17}^2}+4\sqrt{4+Y_{17}^2(1-\sqrt{X_{17}^1})}i}}{16}$$

$$\omega_{17}^{13} = \frac{-1+\sqrt{17}+\Lambda_{17}^1\sqrt{X_{17}^1}-2\sqrt{4+Y_{17}^1(1-\sqrt{X_{17}^2})}+2\sqrt{34-2\sqrt{17}+2\Lambda_{17}^1\sqrt{X_{17}^1}-4\sqrt{4+Y_{17}^1(1+\sqrt{X_{17}^2})}i}}{16}$$

$$\omega_{17}^{14} = \frac{-1+\sqrt{17}+\Lambda_{17}^2\sqrt{X_{17}^2}+2\sqrt{4+Y_{17}^2(1+\sqrt{X_{17}^1})}-2\sqrt{34+2\sqrt{17}+2\Lambda_{17}^2\sqrt{X_{17}^2}-4\sqrt{4+Y_{17}^2(1-\sqrt{X_{17}^1})}i}}{16}$$

$$\omega_{17}^{15} = \frac{-1+\sqrt{17}-\Lambda_{17}^1\sqrt{X_{17}^1}+2\sqrt{4+Y_{17}^1(1+\sqrt{X_{17}^2})}-2\sqrt{34-2\sqrt{17}-2\Lambda_{17}^1\sqrt{X_{17}^1}+4\sqrt{4+Y_{17}^1(1-\sqrt{X_{17}^2})}i}}{16}$$

$$\omega_{17}^{16} = \frac{-1+\sqrt{17}+\Lambda_{17}^1\sqrt{X_{17}^1}+2\sqrt{4+Y_{17}^1(1-\sqrt{X_{17}^2})}-2\sqrt{34-2\sqrt{17}+2\Lambda_{17}^1\sqrt{X_{17}^1}+4\sqrt{4+Y_{17}^1(1+\sqrt{X_{17}^2})}i}}{16}$$

$$\omega_{18}^{1} = -\omega_{9}^{5}$$
$$\omega_{18}^{2} = \omega_{9}^{1}$$
$$\omega_{18}^{3} = \omega_{6}^{1}$$
$$\omega_{18}^{4} = \omega_{9}^{2}$$

$$\omega_{18}^{5} = -\omega_{9}^{7}$$
$$\omega_{18}^{6} = \omega_{3}^{1}$$
$$\omega_{18}^{7} = -\omega_{9}^{8}$$
$$\omega_{18}^{8} = \omega_{9}^{4}$$

$$\omega_{18}^{9} = \omega_{2}^{1}$$
$$\omega_{18}^{10} = \omega_{9}^{5}$$
$$\omega_{18}^{11} = -\omega_{9}^{1}$$
$$\omega_{18}^{12} = \omega_{3}^{2}$$

$$\omega_{18}^{13} = -\omega_{9}^{2}$$
$$\omega_{18}^{14} = \omega_{9}^{7}$$
$$\omega_{18}^{15} = \omega_{6}^{5}$$
$$\omega_{18}^{16} = \omega_{9}^{8}$$

$$\omega_{18}^{17} = -\omega_{9}^{4}$$

19.

$$\vec{\mathrm{A}}_{19} = \sqrt[3]{19}\begin{bmatrix} \sqrt[3]{\frac{7+3\sqrt{3}i}{2}} \\ \sqrt[3]{\frac{7-3\sqrt{3}i}{2}} \end{bmatrix}$$

$$\vec{\mathrm{M}}_{19}^{1} = \begin{bmatrix} \frac{37-3\sqrt{3}i}{2} & \frac{37+3\sqrt{3}i}{2} \end{bmatrix}$$

$$\vec{\mathrm{M}}_{19}^{2} = \begin{bmatrix} \frac{-23-17\sqrt{3}i}{2} & \frac{-23+17\sqrt{3}i}{2} \end{bmatrix}$$

$$\vec{\mathrm{M}}_{19}^{3} = \begin{bmatrix} -7+10\sqrt{3}i & -7-10\sqrt{3}i \end{bmatrix}$$

$$\vec{\mathrm{N}}_{19}^{1} = \begin{bmatrix} \frac{3-\sqrt{3}i}{2} & \frac{3+\sqrt{3}i}{2} \end{bmatrix}$$

$$\vec{\mathrm{N}}_{19}^{2} = \begin{bmatrix} \frac{3+\sqrt{3}i}{2} & \frac{3-\sqrt{3}i}{2} \end{bmatrix}$$

$$\vec{\mathrm{N}}_{19}^{3} = \begin{bmatrix} -\sqrt{3}i & \sqrt{3}i \end{bmatrix}$$

$$\vec{\mathrm{B}}_{19}^{1} = 9 \begin{bmatrix} \frac{76+\vec{M}_{19}^{1}\cdot\vec{\mathrm{A}}_{19}+3\sqrt{3}\vec{N}_{19}^{1}\cdot\vec{\mathrm{A}}_{19}i}{2} \\ \frac{76+\vec{M}_{19}^{1}\cdot\vec{\mathrm{A}}_{19}-3\sqrt{3}\vec{N}_{19}^{1}\cdot\vec{\mathrm{A}}_{19}i}{2} \end{bmatrix}$$

$$\vec{\mathrm{B}}_{19}^{2} = 9 \begin{bmatrix} \frac{76+\vec{M}_{19}^{2}\cdot\vec{\mathrm{A}}_{19}+3\sqrt{3}\vec{N}_{19}^{2}\cdot\vec{\mathrm{A}}_{19}i}{2} \\ \frac{76+\vec{M}_{19}^{2}\cdot\vec{\mathrm{A}}_{19}-3\sqrt{3}\vec{N}_{19}^{2}\cdot\vec{\mathrm{A}}_{19}i}{2} \end{bmatrix}$$

$$\vec{\mathrm{B}}_{19}^{3} = 9 \begin{bmatrix} \frac{76+\vec{M}_{19}^{3}\cdot\vec{\mathrm{A}}_{19}+3\sqrt{3}\vec{N}_{19}^{3}\cdot\vec{\mathrm{A}}_{19}i}{2} \\ \frac{76+\vec{M}_{19}^{3}\cdot\vec{\mathrm{A}}_{19}-3\sqrt{3}\vec{N}_{19}^{3}\cdot\vec{\mathrm{A}}_{19}i}{2} \end{bmatrix}$$

$$\omega_{19}^{1} = \frac{-1+\begin{bmatrix}\vec{\Omega}_{3}^{2} & \vec{\Omega}_{3}^{0}\end{bmatrix}\cdot\begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^{2}\end{bmatrix}+3\sqrt{19-\begin{bmatrix}\vec{\Omega}_{3}^{0} & \vec{\Omega}_{3}^{0}\end{bmatrix}\cdot\begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^{3}\end{bmatrix}}i}{18}$$

$$\omega_{19}^{2} = \frac{-1+\begin{bmatrix}\vec{\Omega}_{3}^{0} & \vec{\Omega}_{3}^{0}\end{bmatrix}\cdot\begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^{3}\end{bmatrix}+3\sqrt{19-\begin{bmatrix}\vec{\Omega}_{3}^{1} & \vec{\Omega}_{3}^{0}\end{bmatrix}\cdot\begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^{2}\end{bmatrix}}i}{18}$$

$$\omega_{19}^{3} = \frac{-1+\begin{bmatrix}\vec{\Omega}_{3}^{0} & \vec{\Omega}_{3}^{2}\end{bmatrix}\cdot\begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^{3}\end{bmatrix}+3\sqrt{19-\begin{bmatrix}\vec{\Omega}_{3}^{1} & \vec{\Omega}_{3}^{2}\end{bmatrix}\cdot\begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^{2}\end{bmatrix}}i}{18}$$

$$\omega_{19}^{4} = \frac{-1+\begin{bmatrix}\vec{\Omega}_{3}^{2} & \vec{\Omega}_{3}^{0}\end{bmatrix}\cdot\begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^{1}\end{bmatrix}+3\sqrt{19-\begin{bmatrix}\vec{\Omega}_{3}^{0} & \vec{\Omega}_{3}^{0}\end{bmatrix}\cdot\begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^{3}\end{bmatrix}}i}{18}$$

$$\omega_{19}^{5} = \frac{-1+\begin{bmatrix}\vec{\Omega}_{3}^{0} & \vec{\Omega}_{3}^{1}\end{bmatrix}\cdot\begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^{3}\end{bmatrix}+3\sqrt{19-\begin{bmatrix}\vec{\Omega}_{3}^{1} & \vec{\Omega}_{3}^{1}\end{bmatrix}\cdot\begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^{2}\end{bmatrix}}i}{18}$$

$$\omega_{19}^{6} = \frac{-1+\begin{bmatrix}\vec{\Omega}_{3}^{1} & \vec{\Omega}_{3}^{2}\end{bmatrix}\cdot\begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^{2}\end{bmatrix}+3\sqrt{19-\begin{bmatrix}\vec{\Omega}_{3}^{2} & \vec{\Omega}_{3}^{2}\end{bmatrix}\cdot\begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^{1}\end{bmatrix}}i}{18}$$

$$\omega_{19}^{7} = \frac{-1+\begin{bmatrix}\vec{\Omega}_{3}^{2} & \vec{\Omega}_{3}^{2}\end{bmatrix}\cdot\begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^{1}\end{bmatrix}+3\sqrt{19-\begin{bmatrix}\vec{\Omega}_{3}^{0} & \vec{\Omega}_{3}^{1}\end{bmatrix}\cdot\begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^{3}\end{bmatrix}}i}{18}$$

$$\omega_{19}^{8} = \frac{-1+\begin{bmatrix}\vec{\Omega}_{3}^{2} & \vec{\Omega}_{3}^{1}\end{bmatrix}\cdot\begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^{1}\end{bmatrix}+3\sqrt{19-\begin{bmatrix}\vec{\Omega}_{3}^{0} & \vec{\Omega}_{3}^{2}\end{bmatrix}\cdot\begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^{3}\end{bmatrix}}i}{18}$$

$$\omega_{19}^{9} = \frac{-1 + \begin{bmatrix}\vec{\Omega}_3^1 & \vec{\Omega}_3^1\end{bmatrix} \cdot \begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^2\end{bmatrix} + 3\sqrt{19 - \begin{bmatrix}\vec{\Omega}_3^2 & \vec{\Omega}_3^0\end{bmatrix} \cdot \begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^1\end{bmatrix}}i}{18}$$

$$\omega_{19}^{10} = \frac{-1 + \begin{bmatrix}\vec{\Omega}_3^1 & \vec{\Omega}_3^1\end{bmatrix} \cdot \begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^2\end{bmatrix} - 3\sqrt{19 - \begin{bmatrix}\vec{\Omega}_3^2 & \vec{\Omega}_3^0\end{bmatrix} \cdot \begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^1\end{bmatrix}}i}{18}$$

$$\omega_{19}^{11} = \frac{-1 + \begin{bmatrix}\vec{\Omega}_3^2 & \vec{\Omega}_3^1\end{bmatrix} \cdot \begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^1\end{bmatrix} - 3\sqrt{19 - \begin{bmatrix}\vec{\Omega}_3^0 & \vec{\Omega}_3^2\end{bmatrix} \cdot \begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^3\end{bmatrix}}i}{18}$$

$$\omega_{19}^{12} = \frac{-1 + \begin{bmatrix}\vec{\Omega}_3^2 & \vec{\Omega}_3^2\end{bmatrix} \cdot \begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^1\end{bmatrix} - 3\sqrt{19 - \begin{bmatrix}\vec{\Omega}_3^0 & \vec{\Omega}_3^1\end{bmatrix} \cdot \begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^3\end{bmatrix}}i}{18}$$

$$\omega_{19}^{13} = \frac{-1 + \begin{bmatrix}\vec{\Omega}_3^1 & \vec{\Omega}_3^2\end{bmatrix} \cdot \begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^2\end{bmatrix} - 3\sqrt{19 - \begin{bmatrix}\vec{\Omega}_3^2 & \vec{\Omega}_3^2\end{bmatrix} \cdot \begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^1\end{bmatrix}}i}{18}$$

$$\omega_{19}^{14} = \frac{-1 + \begin{bmatrix}\vec{\Omega}_3^0 & \vec{\Omega}_3^1\end{bmatrix} \cdot \begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^3\end{bmatrix} - 3\sqrt{19 - \begin{bmatrix}\vec{\Omega}_3^1 & \vec{\Omega}_3^1\end{bmatrix} \cdot \begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^2\end{bmatrix}}i}{18}$$

$$\omega_{19}^{15} = \frac{-1 + \begin{bmatrix}\vec{\Omega}_3^2 & \vec{\Omega}_3^0\end{bmatrix} \cdot \begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^1\end{bmatrix} - 3\sqrt{19 - \begin{bmatrix}\vec{\Omega}_3^0 & \vec{\Omega}_3^0\end{bmatrix} \cdot \begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^3\end{bmatrix}}i}{18}$$

$$\omega_{19}^{16} = \frac{-1 + \begin{bmatrix}\vec{\Omega}_3^0 & \vec{\Omega}_3^2\end{bmatrix} \cdot \begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^3\end{bmatrix} - 3\sqrt{19 - \begin{bmatrix}\vec{\Omega}_3^1 & \vec{\Omega}_3^2\end{bmatrix} \cdot \begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^2\end{bmatrix}}i}{18}$$

$$\omega_{19}^{17} = \frac{-1 + \begin{bmatrix}\vec{\Omega}_3^0 & \vec{\Omega}_3^0\end{bmatrix} \cdot \begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^3\end{bmatrix} - 3\sqrt{19 - \begin{bmatrix}\vec{\Omega}_3^1 & \vec{\Omega}_3^0\end{bmatrix} \cdot \begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^2\end{bmatrix}}i}{18}$$

$$\omega_{19}^{18} = \frac{-1 + \begin{bmatrix}\vec{\Omega}_3^2 & \vec{\Omega}_3^0\end{bmatrix} \cdot \begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^2\end{bmatrix} - 3\sqrt{19 - \begin{bmatrix}\vec{\Omega}_3^0 & \vec{\Omega}_3^0\end{bmatrix} \cdot \begin{bmatrix}\vec{\mathrm{A}}_{19} \\ \vec{\mathrm{B}}_{19}^3\end{bmatrix}}i}{18}$$

20.

$$\omega_{20}^{1} = \frac{\sqrt{10 + 2\sqrt{5}} + (-1 + \sqrt{5})i}{4}$$

$$\omega_{20}^{2} = \omega_{10}^{1}$$

$$\omega_{20}^{3} = \frac{\sqrt{10 - 2\sqrt{5}} + (1 + \sqrt{5})i}{4}$$

$$\omega_{20}^{4} = \omega_{5}^{1}$$

$$\omega_{20}^{5} = \omega_{4}^{1}$$
$$\omega_{20}^{6} = \omega_{10}^{3}$$

$$\omega_{20}^{7} = \frac{-\sqrt{10-2\sqrt{5}}+(1+\sqrt{5})i}{4}$$
$$\omega_{20}^{8} = \omega_{5}^{2}$$
$$\omega_{20}^{9} = \frac{-\sqrt{10+2\sqrt{5}}+(-1+\sqrt{5})i}{4}$$

$$\omega_{20}^{10} = \omega_{2}^{1}$$
$$\omega_{20}^{11} = \frac{-\sqrt{10+2\sqrt{5}}-(-1+\sqrt{5})i}{4}$$
$$\omega_{20}^{12} = \omega_{5}^{3}$$

$$\omega_{20}^{13} = \frac{-\sqrt{10-2\sqrt{5}}-(1+\sqrt{5})i}{4}$$
$$\omega_{20}^{14} = \omega_{10}^{7}$$
$$\omega_{20}^{15} = \omega_{4}^{3}$$

$$\omega_{20}^{16} = \omega_{5}^{4}$$
$$\omega_{20}^{17} = \frac{\sqrt{10-2\sqrt{5}}-(1+\sqrt{5})i}{4}$$
$$\omega_{20}^{18} = \omega_{10}^{9}$$

$$\omega_{20}^{19} = \frac{\sqrt{10+2\sqrt{5}}-(-1+\sqrt{5})i}{4}$$

1.

$$\omega_{21}^{1} = \sqrt[3]{\omega_{7}^{1}}$$
$$\omega_{21}^{2} = \sqrt[3]{\omega_{7}^{2}}$$
$$\omega_{21}^{3} = \omega_{7}^{1}$$
$$\omega_{21}^{4} = \omega_{3}^{1}\omega_{7}^{6}$$
$$\omega_{21}^{5} = \omega_{3}^{1}\sqrt[3]{\omega_{7}^{5}}$$

$$\omega_{21}^{6} = \omega_{7}^{2}$$
$$\omega_{21}^{7} = \omega_{3}^{1}$$
$$\omega_{21}^{8} = \omega_{3}^{1}\sqrt[3]{\omega_{7}^{1}}$$
$$\omega_{21}^{9} = \omega_{7}^{3}$$
$$\omega_{21}^{10} = \omega_{3}^{1}\omega_{7}^{1}$$

$$\omega_{21}^{11} = \omega_{3}^{2}\omega_{7}^{6}$$

$$\omega_{21}^{12} = \omega_7^4$$
$$\omega_{21}^{13} = \omega_3^2 \sqrt[3]{\omega_7^6}$$
$$\omega_{21}^{14} = \omega_3^2$$
$$\omega_{21}^{15} = \omega_7^5$$

$$\omega_{21}^{16} = \omega_3^2 \sqrt[3]{\omega_7^2}$$
$$\omega_{21}^{17} = \omega_3^2 \omega_7^1$$
$$\omega_{21}^{18} = \omega_7^6$$
$$\omega_{21}^{19} = \sqrt[3]{\omega_7^5}$$
$$\omega_{21}^{20} = \sqrt[3]{\omega_7^6}$$

2.

$$\omega_{22}^{1} = -\omega_{11}^{6}$$
$$\omega_{22}^{2} = \omega_{11}^{1}$$
$$\omega_{22}^{3} = -\omega_{11}^{7}$$
$$\omega_{22}^{4} = \omega_{11}^{2}$$

$$\omega_{22}^{5} = -\omega_{11}^{8}$$
$$\omega_{22}^{6} = \omega_{11}^{3}$$
$$\omega_{22}^{7} = -\omega_{11}^{9}$$
$$\omega_{22}^{8} = \omega_{11}^{4}$$

$$\omega_{22}^{9} = -\omega_{11}^{10}$$
$$\omega_{22}^{10} = \omega_{11}^{5}$$
$$\omega_{22}^{11} = \omega_{2}^{1}$$
$$\omega_{22}^{12} = \omega_{11}^{6}$$

$$\omega_{22}^{13} = -\omega_{11}^{1}$$
$$\omega_{22}^{14} = \omega_{11}^{7}$$
$$\omega_{22}^{15} = -\omega_{11}^{2}$$
$$\omega_{22}^{16} = \omega_{11}^{8}$$

$$\omega_{22}^{17} = -\omega_{11}^{3}$$
$$\omega_{22}^{18} = \omega_{11}^{9}$$
$$\omega_{22}^{19} = -\omega_{11}^{4}$$
$$\omega_{22}^{20} = \omega_{11}^{10}$$

$$\omega_{22}^{21} = -\omega_{11}^{5}$$

3.

$$\mathrm{O}_{23}^{1}=\frac{-40957+1443\sqrt{5}+(8004+2627\sqrt{5})\sqrt{10-2\sqrt{5}}i}{4}$$

$$\mathrm{O}_{23}^{2}=\frac{-40957-1443\sqrt{5}+(8004-2627\sqrt{5})\sqrt{10+2\sqrt{5}}i}{4}$$

$$\Pi_{23}^{1}=\frac{750022-112738\sqrt{5}-(57187+9496\sqrt{5})\sqrt{10+2\sqrt{5}}i}{8}$$

$$\Pi_{23}^{2}=\frac{750022+112738\sqrt{5}+(57187-9496\sqrt{5})\sqrt{10-2\sqrt{5}}i}{8}$$

$$\mathrm{P}_{23}^{1}=\frac{-6+10\sqrt{5}-(5-\sqrt{5})\sqrt{10+2\sqrt{5}}i}{8}$$

$$\mathrm{P}_{23}^{2}=\frac{-6-10\sqrt{5}+(5+\sqrt{5})\sqrt{10-2\sqrt{5}}i}{8}$$

$$\boldsymbol{E}_{23}^{1}=\begin{bmatrix}\mathrm{O}_{23}^{1}&0&0&0\\0&\bar{\mathrm{O}}_{23}^{1}&0&0\\0&0&\mathrm{O}_{23}^{2}&0\\0&0&0&\bar{\mathrm{O}}_{23}^{2}\end{bmatrix}$$

$$\boldsymbol{E}_{23}^{2}=\begin{bmatrix}\Pi_{23}^{1}&0&0&0\\0&\bar{\Pi}_{23}^{1}&0&0\\0&0&\Pi_{23}^{2}&0\\0&0&0&\bar{\Pi}_{23}^{2}\end{bmatrix}$$

$$\boldsymbol{E}_{23}^{3}=\begin{bmatrix}\mathrm{P}_{23}^{1}&0&0&0\\0&\bar{\mathrm{P}}_{23}^{1}&0&0\\0&0&\mathrm{P}_{23}^{2}&0\\0&0&0&\bar{\mathrm{P}}_{23}^{2}\end{bmatrix}$$

$$X_{23}^{1}=\frac{11503915+605\vec{\Omega}_{5}^{0}\cdot\boldsymbol{E}_{23}^{1}\cdot\vec{\mathrm{A}}_{11}+(9175606+11\vec{\Omega}_{5}^{0}\cdot\boldsymbol{E}_{23}^{2}\cdot\vec{\mathrm{A}}_{11})\sqrt{110+5\vec{\Omega}_{5}^{2}\cdot\boldsymbol{E}_{23}^{3}\cdot\vec{\mathrm{A}}_{11}}i}{20}$$

$$X_{23}^{2}=\frac{11503915+605\vec{\Omega}_{5}^{1}\cdot\boldsymbol{E}_{23}^{1}\cdot\vec{\mathrm{A}}_{11}+(9175606+11\vec{\Omega}_{5}^{1}\cdot\boldsymbol{E}_{23}^{2}\cdot\vec{\mathrm{A}}_{11})\sqrt{110+5\vec{\Omega}_{5}^{3}\cdot\boldsymbol{E}_{23}^{3}\cdot\vec{\mathrm{A}}_{11}}i}{20}$$

$$X_{23}^{3}=\frac{11503915+605\vec{\Omega}_{5}^{2}\cdot\boldsymbol{E}_{23}^{1}\cdot\vec{\mathrm{A}}_{11}+(9175606+11\vec{\Omega}_{5}^{2}\cdot\boldsymbol{E}_{23}^{2}\cdot\vec{\mathrm{A}}_{11})\sqrt{110+5\vec{\Omega}_{5}^{1}\cdot\boldsymbol{E}_{23}^{3}\cdot\vec{\mathrm{A}}_{11}}i}{20}$$

$$X_{23}^{4}=\frac{11503915+605\vec{\Omega}_{5}^{3}\cdot\boldsymbol{E}_{23}^{1}\cdot\vec{\mathrm{A}}_{11}+(9175606+11\vec{\Omega}_{5}^{3}\cdot\boldsymbol{E}_{23}^{2}\cdot\vec{\mathrm{A}}_{11})\sqrt{110+5\vec{\Omega}_{5}^{4}\cdot\boldsymbol{E}_{23}^{3}\cdot\vec{\mathrm{A}}_{11}}i}{20}$$

$$X_{23}^{5}=\frac{11503915+605\vec{\Omega}_{5}^{4}\cdot\boldsymbol{E}_{23}^{1}\cdot\vec{\mathrm{A}}_{11}+(9175606+11\vec{\Omega}_{5}^{4}\cdot\boldsymbol{E}_{23}^{2}\cdot\vec{\mathrm{A}}_{11})\sqrt{110+5\vec{\Omega}_{5}^{0}\cdot\boldsymbol{E}_{23}^{3}\cdot\vec{\mathrm{A}}_{11}}i}{20}$$

$$\vec{\mathrm{A}}_{23} = \sqrt[11]{23} \begin{bmatrix} \sqrt[11]{X_{23}^{1}} \\ \sqrt[11]{X_{23}^{1}} \\ \sqrt[11]{X_{23}^{2}} \\ \sqrt[11]{X_{23}^{2}} \\ \sqrt[11]{X_{23}^{3}} \\ \sqrt[11]{X_{23}^{3}} \\ \sqrt[11]{X_{23}^{4}} \\ \sqrt[11]{X_{23}^{4}} \\ \sqrt[11]{X_{23}^{5}} \\ \sqrt[11]{X_{23}^{5}} \end{bmatrix}$$

$$\overrightarrow{\Omega}_{11}^{0} = \begin{bmatrix} 1 & 1 & \omega_{11}^{10} & \omega_{11}^{1} & \omega_{11}^{1} & \omega_{11}^{10} & \omega_{11}^{7} & \omega_{11}^{4} & \omega_{11}^{3} & \omega_{11}^{8} \end{bmatrix}$$

$$\overrightarrow{\Omega}_{11}^{1} = \begin{bmatrix} \omega_{11}^{1} & \omega_{11}^{10} & \omega_{11}^{7} & \omega_{11}^{4} & \omega_{11}^{6} & \omega_{11}^{5} & \omega_{11}^{5} & \omega_{11}^{6} & \omega_{11}^{8} & \omega_{11}^{3} \end{bmatrix}$$

$$\overrightarrow{\Omega}_{11}^{2} = \begin{bmatrix} \omega_{11}^{2} & \omega_{11}^{9} & \omega_{11}^{4} & \omega_{11}^{7} & \omega_{11}^{2} & \omega_{11}^{9} & \omega_{11}^{3} & \omega_{11}^{8} & \omega_{11}^{2} & \omega_{11}^{9} \end{bmatrix}$$

$$\overrightarrow{\Omega}_{11}^{3} = \begin{bmatrix} \omega_{11}^{3} & \omega_{11}^{8} & \omega_{11}^{1} & \omega_{11}^{10} & \omega_{11}^{9} & \omega_{11}^{2} & \omega_{11}^{1} & \omega_{11}^{10} & \omega_{11}^{7} & \omega_{11}^{4} \end{bmatrix}$$

$$\overrightarrow{\Omega}_{11}^{4} = \begin{bmatrix} \omega_{11}^{4} & \omega_{11}^{7} & \omega_{11}^{9} & \omega_{11}^{2} & \omega_{11}^{5} & \omega_{11}^{6} & \omega_{11}^{10} & \omega_{11}^{1} & \omega_{11}^{1} & \omega_{11}^{10} \end{bmatrix}$$

$$\overrightarrow{\Omega}_{11}^{5} = \begin{bmatrix} \omega_{11}^{5} & \omega_{11}^{6} & \omega_{11}^{6} & \omega_{11}^{5} & \omega_{11}^{1} & \omega_{11}^{10} & \omega_{11}^{8} & \omega_{11}^{3} & \omega_{11}^{6} & \omega_{11}^{5} \end{bmatrix}$$

$$\overrightarrow{\Omega}_{11}^{6} = \begin{bmatrix} \omega_{11}^{6} & \omega_{11}^{5} & \omega_{11}^{3} & \omega_{11}^{8} & \omega_{11}^{8} & \omega_{11}^{3} & \omega_{11}^{6} & \omega_{11}^{5} & 1 & 1 \end{bmatrix}$$

$$\overrightarrow{\Omega}_{11}^{7} = \begin{bmatrix} \omega_{11}^{7} & \omega_{11}^{4} & 1 & 1 & \omega_{11}^{4} & \omega_{11}^{7} & \omega_{11}^{4} & \omega_{11}^{7} & \omega_{11}^{5} & \omega_{11}^{6} \end{bmatrix}$$

$$\overrightarrow{\Omega}_{11}^{8} = \begin{bmatrix} \omega_{11}^{8} & \omega_{11}^{3} & \omega_{11}^{8} & \omega_{11}^{3} & 1 & 1 & \omega_{11}^{2} & \omega_{11}^{9} & \omega_{11}^{10} & \omega_{11}^{1} \end{bmatrix}$$

$$\overrightarrow{\Omega}_{11}^{9} = \begin{bmatrix} \omega_{11}^{9} & \omega_{11}^{2} & \omega_{11}^{5} & \omega_{11}^{6} & \omega_{11}^{7} & \omega_{11}^{4} & 1 & 1 & \omega_{11}^{4} & \omega_{11}^{7} \end{bmatrix}$$

$$\overrightarrow{\Omega}_{11}^{10} = \begin{bmatrix} \omega_{11}^{10} & \omega_{11}^{1} & \omega_{11}^{2} & \omega_{11}^{9} & \omega_{11}^{3} & \omega_{11}^{8} & \omega_{11}^{9} & \omega_{11}^{2} & \omega_{11}^{9} & \omega_{11}^{2} \end{bmatrix}$$

$$\omega_{23}^{1} = \frac{-1 + \vec{\Omega}_{11}^{8} \cdot \vec{\mathrm{A}}_{23} + \sqrt{253 - 11\vec{\Omega}_{11}^{0} \cdot \vec{\mathrm{A}}_{23}}i}{22}$$

$$\omega_{23}^{2} = \frac{-1 + \vec{\Omega}_{11}^{0} \cdot \vec{\mathrm{A}}_{23} + \sqrt{253 - 11\vec{\Omega}_{11}^{3} \cdot \vec{\mathrm{A}}_{23}}i}{22}$$

$$\omega_{23}^{3} = \frac{-1 + \vec{\Omega}_{11}^{10} \cdot \vec{\mathrm{A}}_{23} + \sqrt{253 - 11\vec{\Omega}_{11}^{7} \cdot \vec{\mathrm{A}}_{23}}i}{22}$$

$$\omega_{23}^{4} = \frac{-1 + \vec{\Omega}_{11}^{3} \cdot \vec{\mathrm{A}}_{23} + \sqrt{253 - 11\vec{\Omega}_{11}^{6} \cdot \vec{\mathrm{A}}_{23}}i}{22}$$

$$\omega_{23}^{5} = \frac{-1 + \vec{\Omega}_{11}^{4} \cdot \vec{\mathrm{A}}_{23} + \sqrt{253 - 11\vec{\Omega}_{11}^{7} \cdot \vec{\mathrm{A}}_{23}}i}{22}$$

$$\omega_{23}^{6} = \frac{-1 + \vec{\Omega}_{11}^{2} \cdot \vec{\mathrm{A}}_{23} + \sqrt{253 - 11\vec{\Omega}_{11}^{5} \cdot \vec{\mathrm{A}}_{23}}i}{22}$$

$$\omega_{23}^{7} = \frac{-1 + \vec{\Omega}_{11}^{9} \cdot \vec{\mathrm{A}}_{23} + \sqrt{253 - 11\vec{\Omega}_{11}^{1} \cdot \vec{\mathrm{A}}_{23}}i}{22}$$

$$\omega_{23}^{8} = \frac{-1 + \vec{\Omega}_{11}^{6} \cdot \vec{\mathrm{A}}_{23} + \sqrt{253 - 11\vec{\Omega}_{11}^{9} \cdot \vec{\mathrm{A}}_{23}}i}{22}$$

$$\omega_{23}^{9} = \frac{-1 + \vec{\Omega}_{11}^{1} \cdot \vec{\mathrm{A}}_{23} + \sqrt{253 - 11\vec{\Omega}_{11}^{4} \cdot \vec{\mathrm{A}}_{23}}i}{22}$$

$$\omega_{23}^{10} = \frac{-1 + \vec{\Omega}_{11}^{7} \cdot \vec{\mathrm{A}}_{23} + \sqrt{253 - 11\vec{\Omega}_{11}^{10} \cdot \vec{\mathrm{A}}_{23}}i}{22}$$

$$\omega_{23}^{11} = \frac{-1 + \vec{\Omega}_{11}^{5} \cdot \vec{\mathrm{A}}_{23} + \sqrt{253 - 11\vec{\Omega}_{11}^{8} \cdot \vec{\mathrm{A}}_{23}}i}{22}$$

$$\omega_{23}^{12} = \frac{-1 + \vec{\Omega}_{11}^{5} \cdot \vec{\mathrm{A}}_{23} - \sqrt{253 - 11\vec{\Omega}_{11}^{8} \cdot \vec{\mathrm{A}}_{23}}i}{22}$$

$$\omega_{23}^{13} = \frac{-1 + \vec{\Omega}_{11}^{7} \cdot \vec{\mathrm{A}}_{23} - \sqrt{253 - 11\vec{\Omega}_{11}^{10} \cdot \vec{\mathrm{A}}_{23}}i}{22}$$

$$\omega_{23}^{14} = \frac{-1 + \vec{\Omega}_{11}^{1} \cdot \vec{\mathrm{A}}_{23} - \sqrt{253 - 11\vec{\Omega}_{11}^{4} \cdot \vec{\mathrm{A}}_{23}}i}{22}$$

$$\omega_{23}^{15} = \frac{-1 + \vec{\Omega}_{11}^{6} \cdot \vec{\mathrm{A}}_{23} - \sqrt{253 - 11\vec{\Omega}_{11}^{9} \cdot \vec{\mathrm{A}}_{23}}i}{22}$$

$$\omega_{23}^{16} = \frac{-1 + \vec{\Omega}_{11}^{9} \cdot \vec{\mathrm{A}}_{23} - \sqrt{253 - 11\vec{\Omega}_{11}^{1} \cdot \vec{\mathrm{A}}_{23}}i}{22}$$

$$\omega_{23}^{17} = \frac{-1 + \vec{\Omega}_{11}^{2} \cdot \vec{\mathrm{A}}_{23} - \sqrt{253 - 11\vec{\Omega}_{11}^{5} \cdot \vec{\mathrm{A}}_{23}}i}{22}$$

$$\omega_{23}^{18} = \frac{-1 + \vec{\Omega}_{11}^{4} \cdot \vec{\mathrm{A}}_{23} - \sqrt{253 - 11\vec{\Omega}_{11}^{7} \cdot \vec{\mathrm{A}}_{23}}i}{22}$$

$$\omega_{23}^{19} = \frac{-1 + \vec{\Omega}_{11}^{3} \cdot \vec{\mathrm{A}}_{23} - \sqrt{253 - 11\vec{\Omega}_{11}^{6} \cdot \vec{\mathrm{A}}_{23}}i}{22}$$

$$\omega_{23}^{20} = \frac{-1 + \vec{\Omega}_{11}^{10} \cdot \vec{\mathrm{A}}_{23} - \sqrt{253 - 11\vec{\Omega}_{11}^{7} \cdot \vec{\mathrm{A}}_{23}}i}{22}$$

$$\omega_{23}^{21} = \frac{-1 + \vec{\Omega}_{11}^{0} \cdot \vec{\mathrm{A}}_{23} - \sqrt{253 - 11\vec{\Omega}_{11}^{3} \cdot \vec{\mathrm{A}}_{23}}i}{22}$$

$$\omega_{23}^{22} = \frac{-1 + \vec{\Omega}_{11}^{8} \cdot \vec{\mathrm{A}}_{23} - \sqrt{253 - 11\vec{\Omega}_{11}^{0} \cdot \vec{\mathrm{A}}_{23}}i}{22}$$

4.

$$\omega_{24}^{1} = \frac{\sqrt{6} + \sqrt{2} + \left(\sqrt{6} - \sqrt{2}\right) i}{4}$$

$$\omega_{24}^{2} = \omega_{12}^{1}$$

$$\omega_{24}^{3} = \omega_{8}^{1}$$

$$\omega_{24}^{4} = \omega_{6}^{1}$$

$$\omega_{24}^{5} = \frac{\sqrt{6} - \sqrt{2} + \left(\sqrt{6} + \sqrt{2}\right) i}{4}$$

$$\omega_{24}^{6} = \omega_{4}^{1}$$

$$\omega_{24}^{7} = \frac{-\sqrt{6} + \sqrt{2} + \left(\sqrt{6} + \sqrt{2}\right) i}{4}$$

$$\omega_{24}^{8} = \omega_{3}^{1}$$

$$\omega_{24}^{9} = \omega_{8}^{3}$$

$$\omega_{24}^{10} = \omega_{12}^{5}$$

$$\omega_{24}^{11} = \frac{-\sqrt{6} - \sqrt{2} + \left(\sqrt{6} - \sqrt{2}\right) i}{4}$$

$$\omega_{24}^{12} = \omega_{2}^{1}$$

$$\omega_{24}^{13} = \frac{-\sqrt{6} - \sqrt{2} - \left(\sqrt{6} - \sqrt{2}\right) i}{4}$$

$$\omega_{24}^{14} = \omega_{12}^{7}$$

$$\omega_{24}^{15} = \omega_{8}^{5}$$

$$\omega_{24}^{16} = \omega_{3}^{2}$$

$$\omega_{24}^{17} = \frac{-\sqrt{6}+\sqrt{2}-\left(\sqrt{6}+\sqrt{2}\right)i}{4}$$
$$\omega_{24}^{18} = \omega_4^3$$

$$\omega_{24}^{19} = \frac{\sqrt{6}-\sqrt{2}-\left(\sqrt{6}+\sqrt{2}\right)i}{4}$$
$$\omega_{24}^{20} = \omega_6^5$$
$$\omega_{24}^{21} = \omega_8^7$$

$$\omega_{24}^{22} = \omega_{12}^{11}$$
$$\omega_{24}^{23} = \frac{\sqrt{6}+\sqrt{2}-\left(\sqrt{6}-\sqrt{2}\right)i}{4}$$

5.

$$\omega_{25}^{1} = \sqrt[5]{\omega_5^1}$$
$$\omega_{25}^{2} = \sqrt[5]{\omega_5^2}$$
$$\omega_{25}^{3} = \omega_5^1\sqrt[5]{\omega_5^3}$$
$$\omega_{25}^{4} = \omega_5^1\sqrt[5]{\omega_5^4}$$
$$\omega_{25}^{5} = \omega_5^1$$

$$\omega_{25}^{6} = \omega_5^1\sqrt[5]{\omega_5^1}$$
$$\omega_{25}^{7} = \omega_5^1\sqrt[5]{\omega_5^2}$$
$$\omega_{25}^{8} = \omega_5^2\sqrt[5]{\omega_5^3}$$
$$\omega_{25}^{9} = \omega_5^2\sqrt[5]{\omega_5^4}$$
$$\omega_{25}^{10} = \omega_5^2$$

$$\omega_{25}^{11} = \omega_5^2\sqrt[5]{\omega_5^1}$$
$$\omega_{25}^{12} = \omega_5^2\sqrt[5]{\omega_5^2}$$
$$\omega_{25}^{13} = \omega_5^3\sqrt[5]{\omega_5^3}$$
$$\omega_{25}^{14} = \omega_5^3\sqrt[5]{\omega_5^4}$$
$$\omega_{25}^{15} = \omega_5^3$$

$$\omega_{25}^{16} = \omega_5^3\sqrt[5]{\omega_5^1}$$
$$\omega_{25}^{17} = \omega_5^3\sqrt[5]{\omega_5^2}$$

$$\omega_{25}^{18} = \omega_5^4 \sqrt[5]{\omega_5^3}$$
$$\omega_{25}^{19} = \omega_5^4 \sqrt[5]{\omega_5^4}$$
$$\omega_{25}^{20} = \omega_5^4$$

$$\omega_{25}^{21} = \omega_5^4 \sqrt[5]{\omega_5^1}$$
$$\omega_{25}^{22} = \omega_5^4 \sqrt[5]{\omega_5^2}$$
$$\omega_{25}^{23} = \sqrt[5]{\omega_5^3}$$
$$\omega_{25}^{24} = \sqrt[5]{\omega_5^4}$$

6.

$$\omega_{26}^{1} = -\omega_{13}^{7}$$
$$\omega_{26}^{2} = \omega_{13}^{1}$$
$$\omega_{26}^{3} = -\omega_{13}^{8}$$
$$\omega_{26}^{4} = \omega_{13}^{2}$$
$$\omega_{26}^{5} = -\omega_{13}^{9}$$
$$\omega_{26}^{6} = \omega_{13}^{3}$$

$$\omega_{26}^{7} = -\omega_{13}^{10}$$
$$\omega_{26}^{8} = \omega_{13}^{4}$$
$$\omega_{26}^{9} = -\omega_{13}^{11}$$
$$\omega_{26}^{10} = \omega_{13}^{5}$$
$$\omega_{26}^{11} = -\omega_{13}^{12}$$
$$\omega_{26}^{12} = \omega_{13}^{6}$$

$$\omega_{26}^{13} = \omega_{2}^{1}$$
$$\omega_{26}^{14} = \omega_{13}^{7}$$
$$\omega_{26}^{15} = -\omega_{13}^{1}$$
$$\omega_{26}^{16} = \omega_{13}^{8}$$
$$\omega_{26}^{17} = -\omega_{13}^{2}$$
$$\omega_{26}^{18} = \omega_{13}^{9}$$

$$\omega_{26}^{19} = -\omega_{13}^{3}$$
$$\omega_{26}^{20} = \omega_{13}^{10}$$
$$\omega_{26}^{21} = -\omega_{13}^{4}$$
$$\omega_{26}^{22} = \omega_{13}^{11}$$
$$\omega_{26}^{23} = -\omega_{13}^{5}$$
$$\omega_{26}^{24} = \omega_{13}^{12}$$

$$\omega_{26}^{25} = -\omega_{13}^{6}$$

7.

$$\omega_{27}^{1} = \sqrt[3]{\omega_{9}^{1}}$$

$$\omega_{27}^{2} = \sqrt[3]{\omega_{9}^{2}}$$

$$\omega_{27}^{3} = \omega_{9}^{1}$$

$$\omega_{27}^{4} = \sqrt[3]{\omega_{9}^{4}}$$

$$\omega_{27}^{5} = \omega_{3}^{1}\sqrt[3]{\omega_{9}^{5}}$$

$$\omega_{27}^{6} = \omega_{9}^{2}$$

$$\omega_{27}^{7} = \omega_{3}^{1}\sqrt[3]{\omega_{9}^{7}}$$

$$\omega_{27}^{8} = \omega_{3}^{1}\sqrt[3]{\omega_{9}^{8}}$$

$$\omega_{27}^{9} = \omega_{3}^{1}$$

$$\omega_{27}^{10} = \omega_{3}^{1}\sqrt[3]{\omega_{3}^{1}}$$

$$\omega_{27}^{11} = \omega_{3}^{1}\sqrt[3]{\omega_{9}^{2}}$$

$$\omega_{27}^{12} = \omega_{9}^{4}$$

$$\omega_{27}^{13} = \omega_{3}^{1}\sqrt[3]{\omega_{9}^{4}}$$

$$\omega_{27}^{14} = \omega_{3}^{2}\sqrt[3]{\omega_{9}^{5}}$$

$$\omega_{27}^{15} = \omega_{9}^{5}$$

$$\omega_{27}^{16} = \omega_{3}^{2}\sqrt[3]{\omega_{9}^{7}}$$

$$\omega_{27}^{17} = \omega_{3}^{2}\sqrt[3]{\omega_{9}^{8}}$$

$$\omega_{27}^{18} = \omega_{3}^{2}$$

$$\omega_{27}^{19} = \omega_{3}^{2}\sqrt[3]{\omega_{3}^{1}}$$

$$\omega_{27}^{20} = \omega_{3}^{2}\sqrt[3]{\omega_{9}^{2}}$$

$$\omega_{27}^{21} = \omega_{9}^{7}$$

$$\omega_{27}^{22} = \omega_{3}^{2}\sqrt[3]{\omega_{9}^{4}}$$

$$\omega_{27}^{23} = \sqrt[3]{\omega_{9}^{5}}$$

$$\omega_{27}^{24} = \omega_{9}^{8}$$

$$\omega_{27}^{25} = \sqrt[3]{\omega_{9}^{7}}$$

$$\omega_{27}^{26} = \sqrt[3]{\omega_{9}^{8}}$$

8.

$$\omega_{28}^{1} = \sqrt{\omega_{14}^{1}}$$
$$\omega_{28}^{2} = \omega_{14}^{1}$$
$$\omega_{28}^{3} = \sqrt{\omega_{14}^{3}}$$
$$\omega_{28}^{4} = \omega_{7}^{1}$$

$$\omega_{28}^{5} = \sqrt{\omega_{14}^{5}}$$
$$\omega_{28}^{6} = \omega_{14}^{3}$$
$$\omega_{28}^{7} = \omega_{4}^{1}$$
$$\omega_{28}^{8} = \omega_{7}^{2}$$

$$\omega_{28}^{9} = -\sqrt{\omega_{14}^{9}}$$
$$\omega_{28}^{10} = \omega_{14}^{5}$$
$$\omega_{28}^{11} = -\sqrt{\omega_{14}^{11}}$$
$$\omega_{28}^{12} = \omega_{7}^{3}$$

$$\omega_{28}^{13} = -\sqrt{\omega_{14}^{13}}$$
$$\omega_{28}^{14} = \omega_{2}^{1}$$
$$\omega_{28}^{15} = -\sqrt{\omega_{14}^{1}}$$
$$\omega_{28}^{16} = \omega_{7}^{4}$$

$$\omega_{28}^{17} = -\sqrt{\omega_{14}^{3}}$$
$$\omega_{28}^{18} = \omega_{14}^{9}$$
$$\omega_{28}^{19} = -\sqrt{\omega_{14}^{5}}$$
$$\omega_{28}^{20} = \omega_{7}^{5}$$

$$\omega_{28}^{21} = \omega_{4}^{3}$$
$$\omega_{28}^{22} = \omega_{14}^{11}$$
$$\omega_{28}^{23} = -\sqrt{\omega_{14}^{9}}$$
$$\omega_{28}^{24} = \omega_{7}^{6}$$

$$\omega_{28}^{25} = -\sqrt{\omega_{14}^{11}}$$
$$\omega_{28}^{26} = \omega_{14}^{13}$$
$$\omega_{28}^{27} = -\sqrt{\omega_{14}^{13}}$$

9.

$$\vec{\Xi}_{29}^{1} = \begin{bmatrix} 86 - 33\sqrt{3}i & 86 + 33\sqrt{3}i \end{bmatrix}$$

$$\vec{\Xi}_{29}^{2} = \begin{bmatrix} \frac{13+119\sqrt{3}i}{2} & \frac{13-119\sqrt{3}i}{2} \end{bmatrix}$$

$$\vec{\Xi}_{29}^{3} = \begin{bmatrix} \frac{-185-53\sqrt{3}i}{2} & \frac{-185+53\sqrt{3}i}{2} \end{bmatrix}$$

$$\vec{\mathrm{H}}_{29}^{1} = \begin{bmatrix} \frac{-523-323\sqrt{3}i}{2} & \frac{-523+323\sqrt{3}i}{2} \end{bmatrix}$$

$$\vec{\mathrm{H}}_{29}^{2} = \begin{bmatrix} 373 - 50\sqrt{3}i & 373 + 50\sqrt{3}i \end{bmatrix}$$

$$\vec{\mathrm{H}}_{29}^{3} = \begin{bmatrix} \frac{-223+423\sqrt{3}i}{2} & \frac{-223-423\sqrt{3}i}{2} \end{bmatrix}$$

$$\vec{\mathrm{Z}}_{29}^{1} = \begin{bmatrix} 4 - \sqrt{3}i & 4 + \sqrt{3}i \end{bmatrix}$$

$$\vec{\mathrm{Z}}_{29}^{2} = \begin{bmatrix} \frac{-1+5\sqrt{3}i}{2} & \frac{-1-5\sqrt{3}i}{2} \end{bmatrix}$$

$$\vec{\mathrm{Z}}_{29}^{3} = \begin{bmatrix} \frac{-7-3\sqrt{3}i}{2} & \frac{-7+3\sqrt{3}i}{2} \end{bmatrix}$$

$$X_{29}^{1} = \frac{-32883 + 147\vec{\Xi}_{29}^{1} \cdot \vec{\mathrm{A}}_{7} + \left(10409 + 7\vec{\mathrm{H}}_{29}^{1} \cdot \vec{\mathrm{A}}_{7}\right)\sqrt{42 + 3\vec{\mathrm{H}}_{29}^{1} \cdot \vec{\mathrm{A}}_{7}}i}{18}$$

$$X_{29}^{2} = \frac{-32883 + 147\vec{\Xi}_{29}^{2} \cdot \vec{\mathrm{A}}_{7} + \left(10409 + 7\vec{\mathrm{H}}_{29}^{2} \cdot \vec{\mathrm{A}}_{7}\right)\sqrt{42 + 3\vec{\mathrm{H}}_{29}^{2} \cdot \vec{\mathrm{A}}_{7}}i}{18}$$

$$X_{29}^{3} = \frac{-32883 + 147\vec{\Xi}_{29}^{3} \cdot \vec{\mathrm{A}}_{7} + \left(10409 + 7\vec{\mathrm{H}}_{29}^{3} \cdot \vec{\mathrm{A}}_{7}\right)\sqrt{42 + 3\vec{\mathrm{H}}_{29}^{3} \cdot \vec{\mathrm{A}}_{7}}i}{18}$$

$$\vec{\mathrm{A}}_{29} = \sqrt[7]{29} \begin{bmatrix} \sqrt[7]{X_{29}^{1}} \\ \sqrt[7]{\bar{X}_{29}^{1}} \\ \sqrt[7]{X_{29}^{2}} \\ \sqrt[7]{\bar{X}_{29}^{2}} \\ \sqrt[7]{X_{29}^{3}} \\ \sqrt[7]{\bar{X}_{29}^{3}} \end{bmatrix}$$

$$L_{29}^{1} = \frac{3 + 3\vec{\bar{\Lambda}}_{13}^{1} \cdot \vec{\mathrm{A}}_{7} - \left(5 + \vec{\Omega}_{3}^{1} \cdot \vec{\mathrm{A}}_{7}\right)\sqrt{42 + 3\vec{\mathrm{Z}}_{29}^{3} \cdot \vec{\mathrm{A}}_{7}}i}{18}$$

$$L_{29}^{2} = \frac{3 + 3\vec{\bar{\Lambda}}_{13}^{2} \cdot \vec{\mathrm{A}}_{7} - \left(5 + \vec{\Omega}_{3}^{2} \cdot \vec{\mathrm{A}}_{7}\right)\sqrt{42 + 3\vec{\mathrm{Z}}_{29}^{1} \cdot \vec{\mathrm{A}}_{7}}i}{18}$$

$$L_{29}^{3} = \frac{3 + 3\vec{\bar{\Lambda}}_{13}^{3} \cdot \vec{\mathrm{A}}_{7} - \left(5 + \vec{\Omega}_{3}^{0} \cdot \vec{\mathrm{A}}_{7}\right)\sqrt{42 + 3\vec{\mathrm{Z}}_{29}^{2} \cdot \vec{\mathrm{A}}_{7}}i}{18}$$

$$\mathbf{\Delta}_{29} = \begin{bmatrix} L_{29}^{1} & 0 & 0 & 0 & 0 & 0 \\ 0 & \bar{L}_{29}^{1} & 0 & 0 & 0 & 0 \\ 0 & 0 & L_{29}^{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \bar{L}_{29}^{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & L_{29}^{3} & 0 \\ 0 & 0 & 0 & 0 & 0 & \bar{L}_{29}^{3} \end{bmatrix}$$

$$\vec{\Omega}_{7}^{0} = \begin{bmatrix} 1 & 1 & \omega_{7}^{4} & \omega_{7}^{3} & 1 & 1 \end{bmatrix}$$

$$\vec{\Omega}_{7}^{1} = \begin{bmatrix} \omega_{7}^{1} & \omega_{7}^{6} & \omega_{7}^{1} & \omega_{7}^{6} & \omega_{7}^{2} & \omega_{7}^{5} \end{bmatrix}$$

$$\vec{\Omega}_{7}^{2} = \begin{bmatrix} \omega_{7}^{2} & \omega_{7}^{5} & \omega_{7}^{5} & \omega_{7}^{2} & \omega_{7}^{4} & \omega_{7}^{3} \end{bmatrix}$$

$$\vec{\Omega}_{7}^{3} = \begin{bmatrix} \omega_{7}^{3} & \omega_{7}^{4} & \omega_{7}^{2} & \omega_{7}^{5} & \omega_{7}^{6} & \omega_{7}^{1} \end{bmatrix}$$

$$\vec{\Omega}_{7}^{4} = \begin{bmatrix} \omega_{7}^{4} & \omega_{7}^{3} & \omega_{7}^{6} & \omega_{7}^{1} & \omega_{7}^{1} & \omega_{7}^{6} \end{bmatrix}$$

$$\vec{\Omega}_{7}^{5} = \begin{bmatrix} \omega_{7}^{5} & \omega_{7}^{2} & \omega_{7}^{3} & \omega_{7}^{4} & \omega_{7}^{3} & \omega_{7}^{4} \end{bmatrix}$$

$$\vec{\Omega}_{7}^{6} = \begin{bmatrix} \omega_{7}^{6} & \omega_{7}^{1} & 1 & 1 & \omega_{7}^{5} & \omega_{7}^{2} \end{bmatrix}$$

30.

$$\omega_{29}^{1} = \frac{-1 + \vec{\Omega}_{7}^{4} \cdot \vec{\mathrm{A}}_{29} + \sqrt{203 + 7\vec{\Omega}_{7}^{1} \cdot \mathbf{\Delta}_{29} \cdot \vec{\mathrm{A}}_{29}} + \sqrt{406 - 14\vec{\Omega}_{7}^{6} \cdot \vec{\mathrm{A}}_{29} - 14\sqrt{203 + 7\vec{\Omega}_{7}^{3} \cdot \mathbf{\Delta} \cdot \vec{\mathrm{A}}_{29}}i}}{28}$$

$$\omega_{29}^{2} = \frac{-1 + \vec{\Omega}_{7}^{6} \cdot \vec{\mathrm{A}}_{29} + \sqrt{203 + 7\vec{\Omega}_{7}^{3} \cdot \mathbf{\Delta}_{29} \cdot \vec{\mathrm{A}}_{29}} + \sqrt{406 - 14\vec{\Omega}_{7}^{1} \cdot \vec{\mathrm{A}}_{29} - 14\sqrt{203 + 7\vec{\Omega}_{7}^{5} \cdot \mathbf{\Delta} \cdot \vec{\mathrm{A}}_{29}}i}}{28}$$

$$\omega_{29}^{3} = \frac{-1 + \vec{\Omega}_{7}^{0} \cdot \vec{\mathrm{A}}_{29} + \sqrt{203 + 7\vec{\Omega}_{7}^{4} \cdot \mathbf{\Delta}_{29} \cdot \vec{\mathrm{A}}_{29}} + \sqrt{406 - 14\vec{\Omega}_{7}^{2} \cdot \vec{\mathrm{A}}_{29} - 14\sqrt{203 + 7\vec{\Omega}_{7}^{6} \cdot \mathbf{\Delta} \cdot \vec{\mathrm{A}}_{29}}i}}{28}$$

$$\omega_{29}^{4} = \frac{-1 + \vec{\Omega}_{7}^{1} \cdot \vec{\mathrm{A}}_{29} + \sqrt{203 + 7\vec{\Omega}_{7}^{5} \cdot \mathbf{\Delta}_{29} \cdot \vec{\mathrm{A}}_{29}} + \sqrt{406 - 14\vec{\Omega}_{7}^{3} \cdot \vec{\mathrm{A}}_{29} - 14\sqrt{203 + 7\vec{\Omega}_{7}^{0} \cdot \mathbf{\Delta} \cdot \vec{\mathrm{A}}_{29}}i}}{28}$$

$$\omega_{29}^{5} = \frac{-1 + \vec{\Omega}_{7}^{6} \cdot \vec{\mathrm{A}}_{29} - \sqrt{203 + 7\vec{\Omega}_{7}^{3} \cdot \mathbf{\Delta}_{29} \cdot \vec{\mathrm{A}}_{29}} + \sqrt{406 - 14\vec{\Omega}_{7}^{1} \cdot \vec{\mathrm{A}}_{29} + 14\sqrt{203 + 7\vec{\Omega}_{7}^{5} \cdot \mathbf{\Delta} \cdot \vec{\mathrm{A}}_{29}}i}}{28}$$

$$\omega_{29}^{6} = \frac{-1 + \vec{\Omega}_{7}^{2} \cdot \vec{\mathrm{A}}_{29} + \sqrt{203 + 7\vec{\Omega}_{7}^{6} \cdot \mathbf{\Delta}_{29} \cdot \vec{\mathrm{A}}_{29}} + \sqrt{406 - 14\vec{\Omega}_{7}^{4} \cdot \vec{\mathrm{A}}_{29} + 14\sqrt{203 + 7\vec{\Omega}_{7}^{1} \cdot \mathbf{\Delta} \cdot \vec{\mathrm{A}}_{29}}i}}{28}$$

$$\omega_{29}^{7} = \frac{-1 + \vec{\Omega}_{7}^{0} \cdot \vec{\mathrm{A}}_{29} - \sqrt{203 + 7\vec{\Omega}_{7}^{4} \cdot \mathbf{\Delta}_{29} \cdot \vec{\mathrm{A}}_{29}} + \sqrt{406 - 14\vec{\Omega}_{7}^{2} \cdot \vec{\mathrm{A}}_{29} + 14\sqrt{203 + 7\vec{\Omega}_{7}^{6} \cdot \mathbf{\Delta} \cdot \vec{\mathrm{A}}_{29}}i}}{28}$$

$$\omega_{29}^{8} = \frac{-1 + \vec{\Omega}_7^3 \cdot \vec{\mathrm{A}}_{29} + \sqrt{203 + 7\vec{\Omega}_7^0 \cdot \boldsymbol{\Delta}_{29} \cdot \vec{\mathrm{A}}_{29}} + \sqrt{406 - 14\vec{\Omega}_7^5 \cdot \vec{\mathrm{A}}_{29} + 14\sqrt{203 + 7\vec{\Omega}_7^4 \cdot \boldsymbol{\Delta} \cdot \vec{\mathrm{A}}_{29}}i}}{28}$$

$$\omega_{29}^{9} = \frac{-1 + \vec{\Omega}_7^3 \cdot \vec{\mathrm{A}}_{29} - \sqrt{203 + 7\vec{\Omega}_7^0 \cdot \boldsymbol{\Delta}_{29} \cdot \vec{\mathrm{A}}_{29}} + \sqrt{406 - 14\vec{\Omega}_7^5 \cdot \vec{\mathrm{A}}_{29} - 14\sqrt{203 + 7\vec{\Omega}_7^4 \cdot \boldsymbol{\Delta} \cdot \vec{\mathrm{A}}_{29}}i}}{28}$$

$$\omega_{29}^{10} = \frac{-1 + \vec{\Omega}_7^1 \cdot \vec{\mathrm{A}}_{29} - \sqrt{203 + 7\vec{\Omega}_7^5 \cdot \boldsymbol{\Delta}_{29} \cdot \vec{\mathrm{A}}_{29}} + \sqrt{406 - 14\vec{\Omega}_7^3 \cdot \vec{\mathrm{A}}_{29} + 14\sqrt{203 + 7\vec{\Omega}_7^0 \cdot \boldsymbol{\Delta} \cdot \vec{\mathrm{A}}_{29}}i}}{28}$$

$$\omega_{29}^{11} = \frac{-1 + \vec{\Omega}_7^5 \cdot \vec{\mathrm{A}}_{29} + \sqrt{203 + 7\vec{\Omega}_7^2 \cdot \boldsymbol{\Delta}_{29} \cdot \vec{\mathrm{A}}_{29}} + \sqrt{406 - 14\vec{\Omega}_7^0 \cdot \vec{\mathrm{A}}_{29} + 14\sqrt{203 + 7\vec{\Omega}_7^4 \cdot \boldsymbol{\Delta} \cdot \vec{\mathrm{A}}_{29}}i}}{28}$$

$$\omega_{29}^{12} = \frac{-1 + \vec{\Omega}_7^4 \cdot \vec{\mathrm{A}}_{29} - \sqrt{203 + 7\vec{\Omega}_7^1 \cdot \boldsymbol{\Delta}_{29} \cdot \vec{\mathrm{A}}_{29}} + \sqrt{406 - 14\vec{\Omega}_7^6 \cdot \vec{\mathrm{A}}_{29} + 14\sqrt{203 + 7\vec{\Omega}_7^3 \cdot \boldsymbol{\Delta} \cdot \vec{\mathrm{A}}_{29}}i}}{28}$$

$$\omega_{29}^{13} = \frac{-1 + \vec{\Omega}_7^5 \cdot \vec{\mathrm{A}}_{29} - \sqrt{203 + 7\vec{\Omega}_7^2 \cdot \boldsymbol{\Delta}_{29} \cdot \vec{\mathrm{A}}_{29}} + \sqrt{406 - 14\vec{\Omega}_7^0 \cdot \vec{\mathrm{A}}_{29} - 14\sqrt{203 + 7\vec{\Omega}_7^4 \cdot \boldsymbol{\Delta} \cdot \vec{\mathrm{A}}_{29}}i}}{28}$$

$$\omega_{29}^{14} = \frac{-1 + \vec{\Omega}_7^2 \cdot \vec{\mathrm{A}}_{29} - \sqrt{203 + 7\vec{\Omega}_7^6 \cdot \boldsymbol{\Delta}_{29} \cdot \vec{\mathrm{A}}_{29}} + \sqrt{406 - 14\vec{\Omega}_7^4 \cdot \vec{\mathrm{A}}_{29} - 14\sqrt{203 + 7\vec{\Omega}_7^1 \cdot \boldsymbol{\Delta} \cdot \vec{\mathrm{A}}_{29}}i}}{28}$$

$$\omega_{29}^{15} = \frac{-1 + \vec{\Omega}_7^2 \cdot \vec{\mathrm{A}}_{29} - \sqrt{203 + 7\vec{\Omega}_7^6 \cdot \boldsymbol{\Delta}_{29} \cdot \vec{\mathrm{A}}_{29}} - \sqrt{406 - 14\vec{\Omega}_7^4 \cdot \vec{\mathrm{A}}_{29} - 14\sqrt{203 + 7\vec{\Omega}_7^1 \cdot \boldsymbol{\Delta} \cdot \vec{\mathrm{A}}_{29}}i}}{28}$$

$$\omega_{29}^{16} = \frac{-1 + \vec{\Omega}_7^5 \cdot \vec{\mathrm{A}}_{29} - \sqrt{203 + 7\vec{\Omega}_7^2 \cdot \boldsymbol{\Delta}_{29} \cdot \vec{\mathrm{A}}_{29}} - \sqrt{406 - 14\vec{\Omega}_7^0 \cdot \vec{\mathrm{A}}_{29} - 14\sqrt{203 + 7\vec{\Omega}_7^4 \cdot \boldsymbol{\Delta} \cdot \vec{\mathrm{A}}_{29}}i}}{28}$$

$$\omega_{29}^{17} = \frac{-1 + \vec{\Omega}_7^4 \cdot \vec{\mathrm{A}}_{29} - \sqrt{203 + 7\vec{\Omega}_7^1 \cdot \boldsymbol{\Delta}_{29} \cdot \vec{\mathrm{A}}_{29}} - \sqrt{406 - 14\vec{\Omega}_7^6 \cdot \vec{\mathrm{A}}_{29} + 14\sqrt{203 + 7\vec{\Omega}_7^3 \cdot \boldsymbol{\Delta} \cdot \vec{\mathrm{A}}_{29}}i}}{28}$$

$$\omega_{29}^{18} = \frac{-1 + \vec{\Omega}_7^5 \cdot \vec{\mathrm{A}}_{29} + \sqrt{203 + 7\vec{\Omega}_7^2 \cdot \boldsymbol{\Delta}_{29} \cdot \vec{\mathrm{A}}_{29}} - \sqrt{406 - 14\vec{\Omega}_7^0 \cdot \vec{\mathrm{A}}_{29} + 14\sqrt{203 + 7\vec{\Omega}_7^4 \cdot \boldsymbol{\Delta} \cdot \vec{\mathrm{A}}_{29}}i}}{28}$$

$$\omega_{29}^{19} = \frac{-1 + \vec{\Omega}_7^1 \cdot \vec{\mathrm{A}}_{29} - \sqrt{203 + 7\vec{\Omega}_7^5 \cdot \boldsymbol{\Delta}_{29} \cdot \vec{\mathrm{A}}_{29}} - \sqrt{406 - 14\vec{\Omega}_7^3 \cdot \vec{\mathrm{A}}_{29} + 14\sqrt{203 + 7\vec{\Omega}_7^0 \cdot \boldsymbol{\Delta} \cdot \vec{\mathrm{A}}_{29}}i}}{28}$$

$$\omega_{29}^{20} = \frac{-1 + \vec{\Omega}_7^3 \cdot \vec{\mathrm{A}}_{29} - \sqrt{203 + 7\vec{\Omega}_7^0 \cdot \boldsymbol{\Delta}_{29} \cdot \vec{\mathrm{A}}_{29}} - \sqrt{406 - 14\vec{\Omega}_7^5 \cdot \vec{\mathrm{A}}_{29} - 14\sqrt{203 + 7\vec{\Omega}_7^4 \cdot \boldsymbol{\Delta} \cdot \vec{\mathrm{A}}_{29}}i}}{28}$$

$$\omega_{29}^{21} = \frac{-1 + \vec{\Omega}_7^3 \cdot \vec{\mathrm{A}}_{29} + \sqrt{203 + 7\vec{\Omega}_7^0 \cdot \boldsymbol{\Delta}_{29} \cdot \vec{\mathrm{A}}_{29}} - \sqrt{406 - 14\vec{\Omega}_7^5 \cdot \vec{\mathrm{A}}_{29} + 14\sqrt{203 + 7\vec{\Omega}_7^4 \cdot \boldsymbol{\Delta} \cdot \vec{\mathrm{A}}_{29}}i}}{28}$$

$$\omega_{29}^{22} = \frac{-1 + \vec{\Omega}_7^0 \cdot \vec{\mathrm{A}}_{29} - \sqrt{203 + 7\vec{\Omega}_7^4 \cdot \boldsymbol{\Delta}_{29} \cdot \vec{\mathrm{A}}_{29}} - \sqrt{406 - 14\vec{\Omega}_7^2 \cdot \vec{\mathrm{A}}_{29} + 14\sqrt{203 + 7\vec{\Omega}_7^6 \cdot \boldsymbol{\Delta} \cdot \vec{\mathrm{A}}_{29}}i}}{28}$$

$$\omega_{29}^{23} = \frac{-1 + \vec{\Omega}_7^2 \cdot \vec{\mathrm{A}}_{29} + \sqrt{203 + 7\vec{\Omega}_7^6 \cdot \boldsymbol{\Delta}_{29} \cdot \vec{\mathrm{A}}_{29}} - \sqrt{406 - 14\vec{\Omega}_7^4 \cdot \vec{\mathrm{A}}_{29} + 14\sqrt{203 + 7\vec{\Omega}_7^1 \cdot \boldsymbol{\Delta} \cdot \vec{\mathrm{A}}_{29}}i}}{28}$$

$$\omega_{29}^{24} = \frac{-1 + \vec{\Omega}_7^6 \cdot \vec{\mathrm{A}}_{29} - \sqrt{203 + 7\vec{\Omega}_7^3 \cdot \boldsymbol{\Delta}_{29} \cdot \vec{\mathrm{A}}_{29}} - \sqrt{406 - 14\vec{\Omega}_7^1 \cdot \vec{\mathrm{A}}_{29} + 14\sqrt{203 + 7\vec{\Omega}_7^5 \cdot \boldsymbol{\Delta} \cdot \vec{\mathrm{A}}_{29}}i}}{28}$$

$$\omega_{29}^{25} = \frac{-1 + \vec{\Omega}_7^1 \cdot \vec{\mathrm{A}}_{29} + \sqrt{203 + 7\vec{\Omega}_7^5 \cdot \boldsymbol{\Delta}_{29} \cdot \vec{\mathrm{A}}_{29}} - \sqrt{406 - 14\vec{\Omega}_7^3 \cdot \vec{\mathrm{A}}_{29} - 14\sqrt{203 + 7\vec{\Omega}_7^0 \cdot \boldsymbol{\Delta} \cdot \vec{\mathrm{A}}_{29}}i}}{28}$$

$$\omega_{29}^{26} = \frac{-1 + \vec{\Omega}_7^0 \cdot \vec{\mathrm{A}}_{29} + \sqrt{203 + 7\vec{\Omega}_7^4 \cdot \boldsymbol{\Delta}_{29} \cdot \vec{\mathrm{A}}_{29}} - \sqrt{406 - 14\vec{\Omega}_7^2 \cdot \vec{\mathrm{A}}_{29} - 14\sqrt{203 + 7\vec{\Omega}_7^6 \cdot \boldsymbol{\Delta} \cdot \vec{\mathrm{A}}_{29}}i}}{28}$$

$$\omega_{29}^{27} = \frac{-1 + \vec{\Omega}_7^6 \cdot \vec{\mathrm{A}}_{29} + \sqrt{203 + 7\vec{\Omega}_7^3 \cdot \boldsymbol{\Delta}_{29} \cdot \vec{\mathrm{A}}_{29}} - \sqrt{406 - 14\vec{\Omega}_7^1 \cdot \vec{\mathrm{A}}_{29} - 14\sqrt{203 + 7\vec{\Omega}_7^5 \cdot \boldsymbol{\Delta} \cdot \vec{\mathrm{A}}_{29}}i}}{28}$$

$$\omega_{29}^{28} = \frac{-1 + \vec{\Omega}_7^4 \cdot \vec{\mathrm{A}}_{29} + \sqrt{203 + 7\vec{\Omega}_7^1 \cdot \boldsymbol{\Delta}_{29} \cdot \vec{\mathrm{A}}_{29}} - \sqrt{406 - 14\vec{\Omega}_7^6 \cdot \vec{\mathrm{A}}_{29} - 14\sqrt{203 + 7\vec{\Omega}_7^3 \cdot \boldsymbol{\Delta} \cdot \vec{\mathrm{A}}_{29}}i}}{28}$$

$$\omega_{30}^{1} = -\omega_{15}^{8}$$
$$\omega_{30}^{2} = \omega_{15}^{1}$$
$$\omega_{30}^{3} = \omega_{10}^{1}$$
$$\omega_{30}^{4} = \omega_{15}^{2}$$
$$\omega_{30}^{5} = \omega_{6}^{1}$$
$$\omega_{30}^{6} = \omega_{5}^{1}$$

$$\omega_{30}^{7} = -\omega_{15}^{11}$$
$$\omega_{30}^{8} = \omega_{15}^{4}$$
$$\omega_{30}^{9} = \omega_{10}^{3}$$
$$\omega_{30}^{10} = \omega_{3}^{1}$$
$$\omega_{30}^{11} = -\omega_{15}^{13}$$
$$\omega_{30}^{12} = \omega_{5}^{2}$$

$$\omega_{30}^{13} = -\omega_{15}^{14}$$
$$\omega_{30}^{14} = \omega_{15}^{7}$$
$$\omega_{30}^{15} = \omega_{2}^{1}$$
$$\omega_{30}^{16} = \omega_{15}^{8}$$
$$\omega_{30}^{17} = -\omega_{15}^{1}$$
$$\omega_{30}^{18} = \omega_{5}^{3}$$

$$\omega_{30}^{19} = -\omega_{15}^{2}$$
$$\omega_{30}^{20} = \omega_{3}^{2}$$
$$\omega_{30}^{21} = \omega_{10}^{7}$$
$$\omega_{30}^{22} = \omega_{15}^{11}$$
$$\omega_{30}^{23} = -\omega_{15}^{4}$$

$$\omega_{30}^{24} = \omega_{5}^{4}$$

$$\omega_{30}^{25} = \omega_{6}^{5}$$
$$\omega_{30}^{26} = \omega_{15}^{13}$$
$$\omega_{30}^{27} = \omega_{10}^{9}$$
$$\omega_{30}^{28} = \omega_{15}^{14}$$
$$\omega_{30}^{29} = -\omega_{15}^{7}$$